

Brian Soh Han Yu

<https://github.com/NAIRBS> | Email: sohhanyu@gmail.com

SKILLS AND EXPERIENCES

Programming: C, C++, C#, .NET, Java, JavaScript, Python, PowerShell/Batch/Bash/Zsh Scripting, HTML/CSS.

Frameworks/Platforms: Desktop/Mobile/Web Applications, **PERN** (PostgreSQL + Express.js/NestJS + React + Node.js)

EDUCATION

Nanyang Technological University, Singapore

Aug 2022 - Dec 2025

Bachelor of Engineering in Computer Engineering

- Elective Focus in Security (Coursework: Computer Security, Malware Analysis and Cyber Threat Intelligence)

Singapore Polytechnic, Singapore

Apr 2017 - Apr 2020

Diploma in Computer Engineering

- Specialize in Computer Applications

AWARDS AND ACHIEVEMENTS

Singtel SHINE Engineering Cadet Programme Scholarship

Mar 2018 - Apr 2020

WORK EXPERIENCE

Singtel, iLab (1 Year Scholarship Bond active, available 11th Jan 2027)

Jan 2026 - Current

Digital Engineer (Network Lab & Test Process)

- Automated mobile Wi-Fi speed tests within an anechoic chamber by programming an Arduino Uno R4 in C (HID mouse emulation) to simulate touch inputs, greatly reducing human labor and improving test efficiency by 30%.
- Built an air-gapped web-to-CLI relay system on a salvaged Odroid C2 micro-server to automate generation and execution of SSH/Telnet commands across 10+ GPON OLTs over an isolated router network.
- Drove increased lab efficiency by maintaining network infrastructure, installing servers, managing hardware inventory and executing UATs to understand tester workflows and translate them into clear automation requirements.
- Led end-to-end modernization of test management systems by coordinating vendor delivery, reverse-engineering legacy database schemas for data migration, integrating automated tests, and documenting manual workflows.

ACADEMIC PROJECTS

Nanyang Technological University, Singapore

Jan 2025 - Dec 2025

Final Year Project – Augmented Reality Headset

- Designed and developed a prototype edge-enabled AR headset on the Jetson Xavier NX (Linux), streaming real-time augmented output to an external display without an integrated head-mounted monitor due to budget constraints.
- Developed a visual localization system based on OV2SLAM and ORB-SLAM2 in C++, incorporating web workers and low-latency optimizations to enable responsive AR experiences on resource-constrained systems.
- Calibrated stereo ESP32 cameras and optimized SLAM for accurate localization, implementing all AR logic in JavaScript and running the C++ SLAM backend via WebAssembly to run real-time AR directly in web browsers.

INTERNSHIP EXPERIENCE

NCS Singapore

Jul 2024 - Dec 2024

Software Engineering/Quality Assurance, Intern

- Developed test cases, conducted comprehensive end-to-end testing, identified software bugs and previously unconsidered edge cases, and collaborated with developers to ensure timely resolution and quality assurance.
- Documented testing procedures, analyzed client requirements, and produced detailed meeting minutes and Software Requirements Specifications (SRS) reports for multiple projects to support development and quality assurance.

NCS Singapore

Mar 2019 - Aug 2019

Software Engineer, SHINE Intern

- Strategically planned and developed automated processes using Batch Scripting and PowerShell to notify hundreds of domain users about password expiration, completed within 2 weeks with management planning and design.
- Developed IT hardware asset inventory management desktop application in .NET and C# to track existing inventory across multiple locations and generate supplier-specific reports, together with system testing and integration testing.

CO-CURRICULAR ACTIVITIES

NTU HPC (High Performance Computing)

Dec 2023 - Dec 2024

Student HPC Engineer / Competition Team Member

- Achieved 2nd Place globally representing NTU's High-Performance Computing team at HPC-AI 2024, optimizing MPI inter-node communication, profiling software stacks, and scaling HOOMD-blue particle simulations across 1,500+ CPU cores on Australia's NCI Gadi supercomputing cluster.